

Algebra: Evaluating Expressions — Set B



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DIRECTIONS

Substitute the given value(s) for each variable, then simplify using the order of operations. Show each substitution step.

1. Evaluate when $x = 6$: $5x - 4$ Answer: _____	2. Evaluate when $t = -3$: $4t + 10$ Answer: _____
3. Evaluate when $m = 4$: $m^2 - 5m + 6$ Answer: _____	4. Evaluate when $c = -2$: $3c^2 + c - 7$ Answer: _____
5. Evaluate when $x=5, y=-3$: $x + 3y$ Answer: _____	6. Evaluate when $x=5, y=-3$: $2x^2 - y$ Answer: _____
7. Evaluate when $a=2, b=-4$: $a^2 - ab + b^2$ Answer: _____	8. Evaluate when $p=-1, q=3$: $4p^2 - 2pq + q^2$ Answer: _____
9. Evaluate when $x=3, y=-2$: $x^2y - xy^2$ Answer: _____	10. Evaluate when $m=2, n=-3$: $m^3 - 2m^2n + n^2$ Answer: _____

Based on the Numberbender lesson "ALGEBRA - Evaluating Algebraic Expressions | how to?" • youtu.be/CjxUTr6FxGc

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Evaluating Expressions, Part 3 Worksheet

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Answer Key & Solutions

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TEACHER NOTES

Set B extends Set A with slightly more complex expressions. Items 1–4: one variable. Items 5–7: two variables including negatives. Items 8–10: multi-term expressions with exponents.

1. Evaluate when $x = 6$: $5x - 4$

Answer: 26

$$5(6) - 4 = 30 - 4 = 26.$$

2. Evaluate when $t = -3$: $4t + 10$

Answer: -2

$$4(-3) + 10 = -12 + 10 = -2.$$

3. Evaluate when $m = 4$: $m^2 - 5m + 6$

Answer: 2

$$16 - 20 + 6 = 2.$$

4. Evaluate when $c = -2$: $3c^2 + c - 7$

Answer: 3

$$3(4) + (-2) - 7 = 12 - 2 - 7 = 3.$$

5. Evaluate when $x=5$, $y=-3$: $x + 3y$

Answer: -4

$$5 + 3(-3) = 5 - 9 = -4.$$

6. Evaluate when $x=5$, $y=-3$: $2x^2 - y$

Answer: 53

$$2(25) - (-3) = 50 + 3 = 53.$$

7. Evaluate when $a=2$, $b=-4$: $a^2 - ab + b^2$

Answer: 28

$$4 - (-8) + 16 = 4 + 8 + 16 = 28.$$

8. Evaluate when $p=-1$, $q=3$: $4p^2 - 2pq + q^2$

Answer: 19

$$4(1) - 2(-1)(3) + (9) = 4 + 6 + 9 = 19.$$

9. Evaluate when $x=3$, $y=-2$: $x^2y - xy^2$

Answer: -30

$$(9)(-2) - (3)(4) = -18 - 12 = -30.$$

10. Evaluate when $m=2$, $n=-3$: $m^3 - 2m^2n + n^2$

Answer: 41

$$8 - 2(4)(-3) + 9 = 8 + 24 + 9 = 41.$$

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